Learning Probabilistic Models

Faculty of Information Technology
Monash University, Australia

Week 12

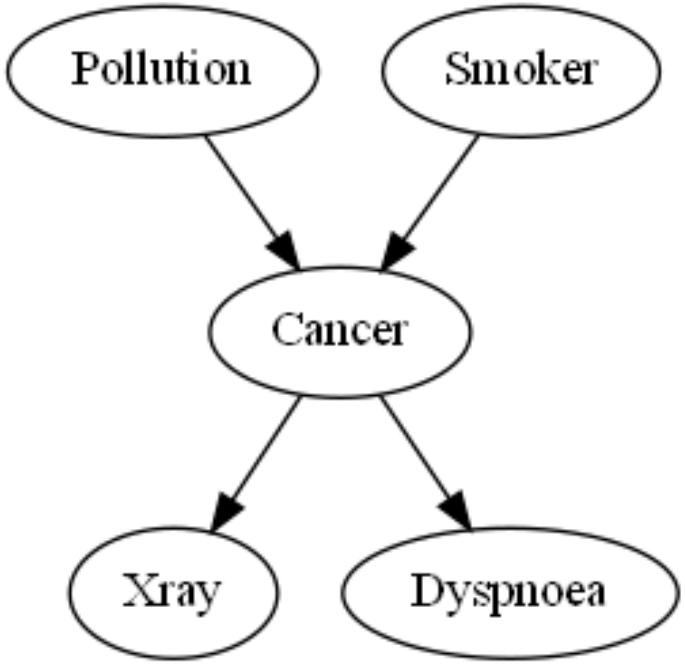
Learning Outcomes

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- After learning this module, you should be able to:
  - Understand the PC algorithm.
  - Apply structure learning algorithms to real-world problems.
- Alignment with unit learning outcomes:
  - Evaluate the limitations, appropriateness and benefits of data analytics methods for given tasks.
  - Design solutions to real-world problems with data analytic techniques.
  - Assess the results of an analysis.

Recap: BN for Lung Cancer

Goal: Learning the structure of a Bayesian network from a dataset?



Random variable	Values
Pollution	low, high
Smoker	True, False
Cancer	True, False
Dyspnoea	True, False
X-ray	pos, neg

Independence Tests

- Goal: given joint samples of two variables X and Y, we want to determine whether X and Y are independent.
- Unconditional independencies.
  - Null hypothesis H_0: $P(X,Y) = P(X)P(Y)$.
  - H_1: $P(X,Y) \neq P(X)P(Y)$
- Conditional independencies:
  - H_0: $P(X,Y|Z) = P(X|Z)P(Y|Z)$
  - H_1: $P(X,Y|Z) \neq P(X|Z)P(Y|Z)$
- Compute a deviation measure from the null hypothesis, such as χ^2 statistic and mutual information.
- The null hypothesis is
  - Accepted, if p-value is larger than a significance level, e.g. 0.05.
  - Rejected, if p-value is smaller than a significance level.

Independence Tests in Practice

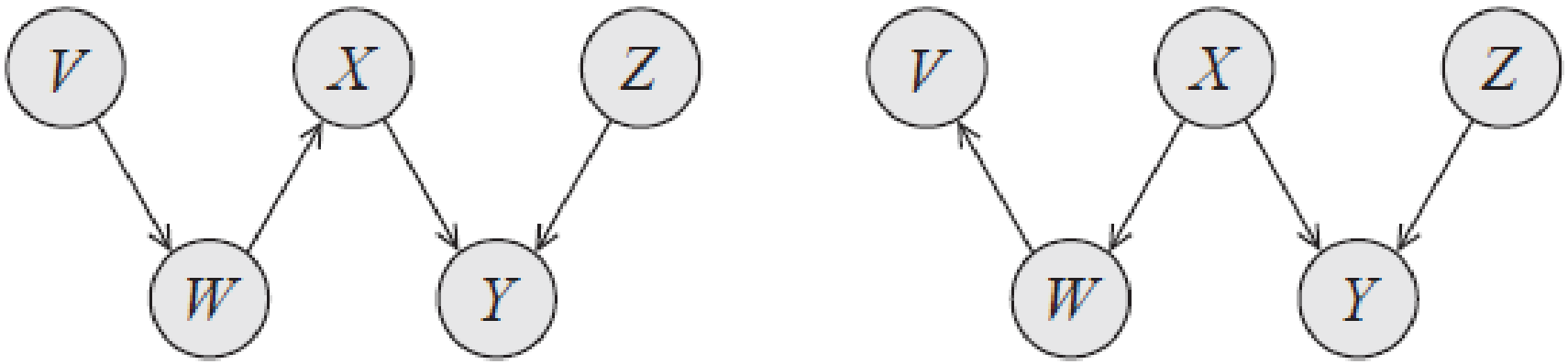
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- χ^2 test:
  - Categorical variables.
  - pgmpy.estimators.CITests.chi_square(X, Y, Z, data)
- Residualization-based CI test:
  - Categorical and ordinal variables (high dimensional)
  - pgmpy.estimators.CITests.pillai_trace(X, Y, Z, data) [ci_pillai() in the early version].
- Pearson correlation based CI test:
  - Continuous variables
  - pgmpy.estimators.CITests.pearsonr(X, Y, Z, data)
- Limitations: sensitive to statistical noise so that the independence test results may be wrong.



I-Equivalence

- $I(G)$ specifies a set of conditional independence assertions that are associated with the graph G.
- I-Equivalence: Two graph structures G_i and G_j over X and are I-equivalent if $I(G_i) = I(G_j)$.

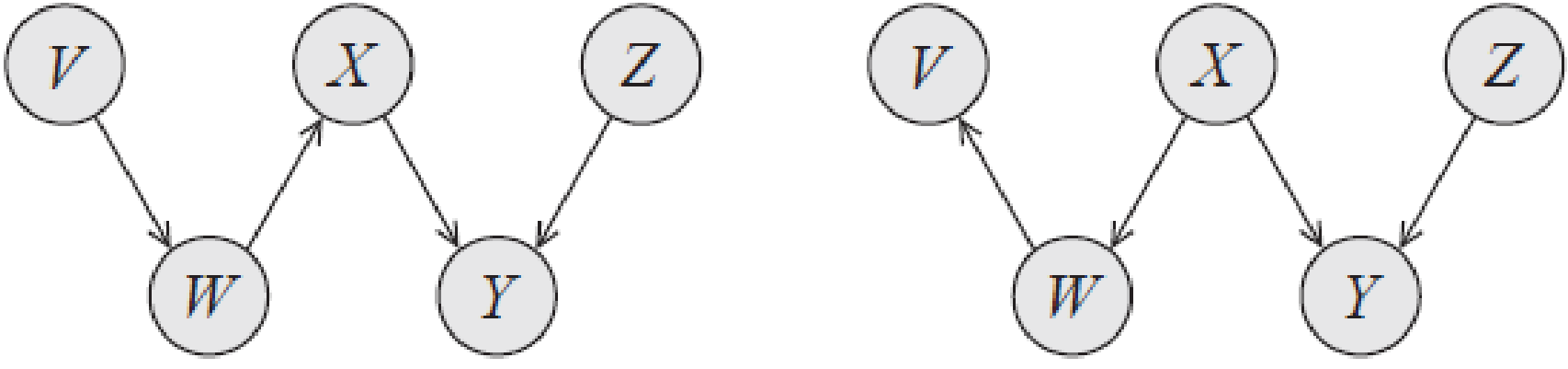


I-equivalence of two graphs implies that any distribution P that can be factorized over of one of these graphs can be factorized over the other.

$$P(Y|X,Z)P(X|W)P(W|V)P(V)P(Z) \Rightarrow P(Y|X,Z)P(W|X)P(V|W)P(X)P(Z)$$

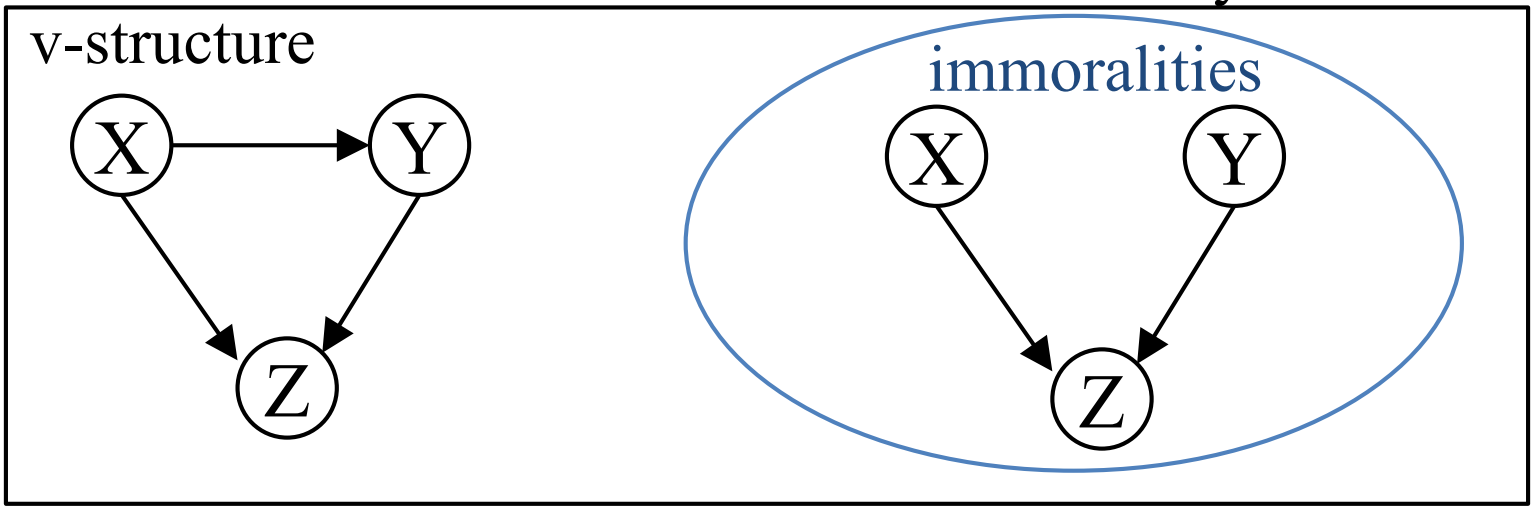
I-Equivalence

- Skeleton: The skeleton of a Bayesian network G over X is an undirected graph over X that contains an edge $X_i - X_j$ for every edge $X_i \rightarrow X_j \; or X_i \leftarrow X_j \; in \; G$.

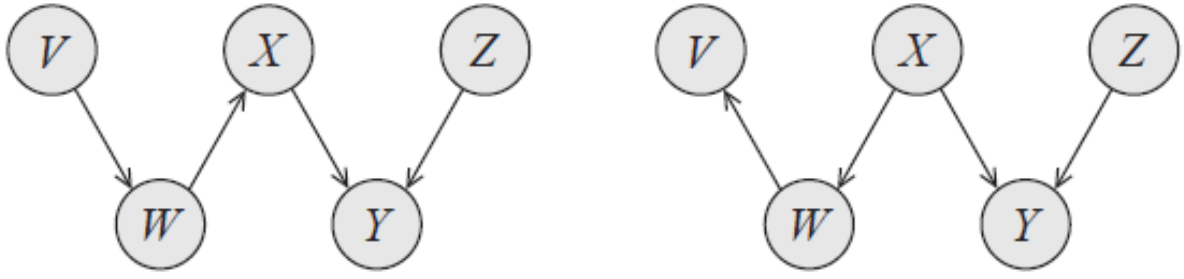


I-Equivalence

- Differences between v-structure and immorality.



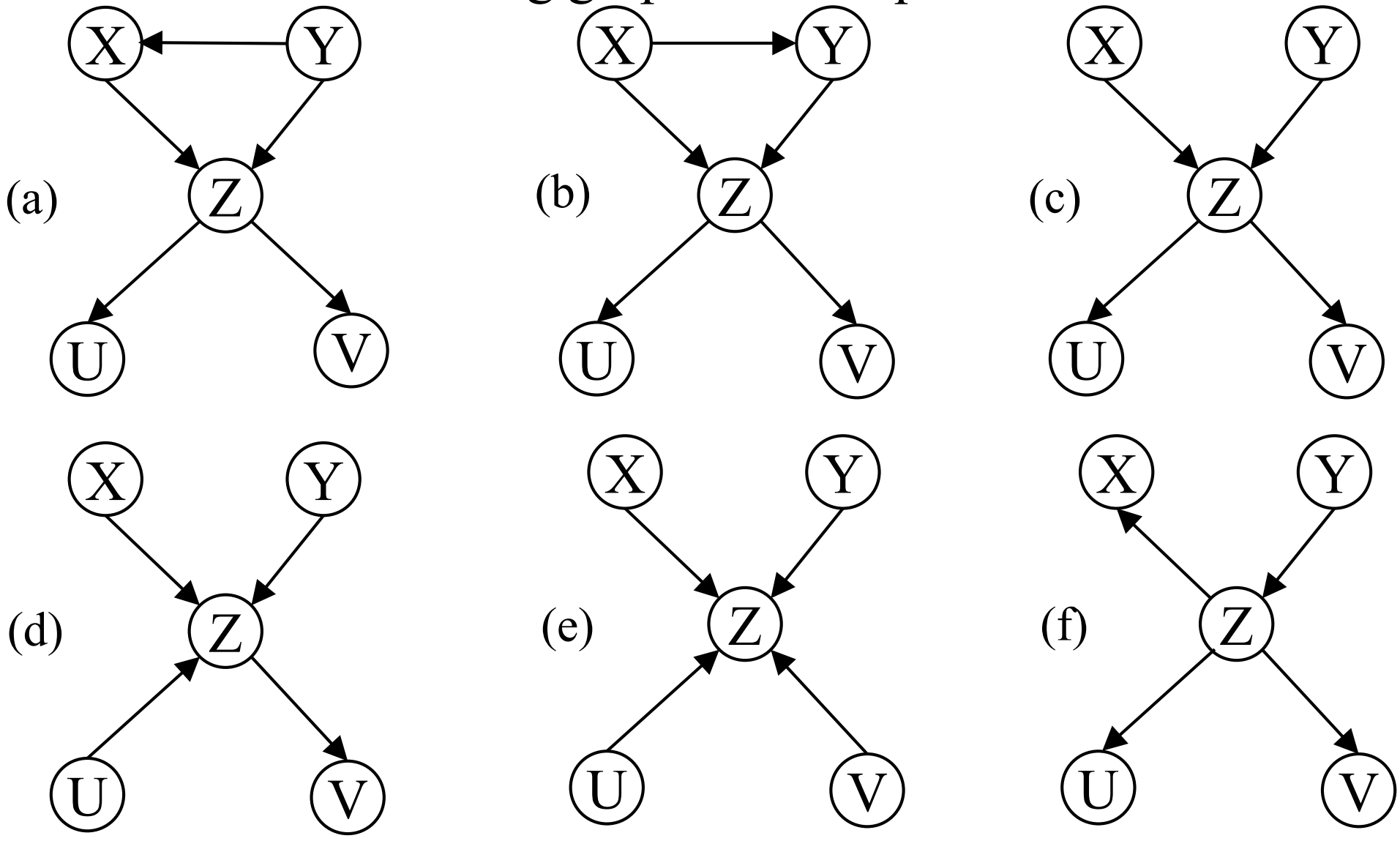
- Two graphs G_i and G_j over X are I-equivalent, if and only if G_i and G_j have the same skeleton and the same set of immoralities.



Question: I-Equivalence

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Which of the following graphs are I-equivalent?



Perfect Maps

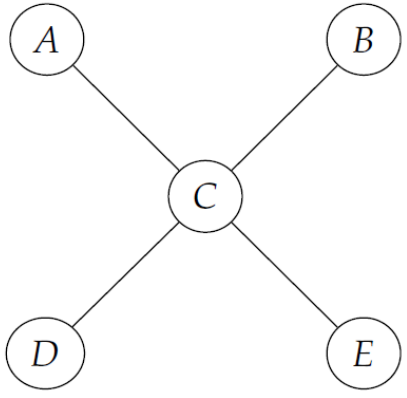
- **Perfect map**: a graph *G* is a perfect map (P-map) for a set of independencies of a distribution P, if *I(G) = I(P)*.
- The P-map of a distribution is generally not unique, but is unique up to I-equivalence between graphs. A distribution P may have many P-Maps, but all of them are I-equivalent.
- Two DAGs are **I-equivalent** if they share
  - the same skeleton and
  - the same set of immoralities.

PC Algorithm

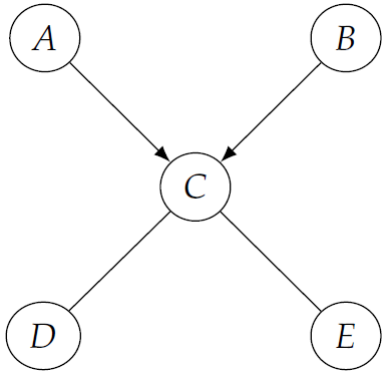
Goal: instead of a single P-map, find an I-equivalence class of a distribution P, and use heuristics to identify the orientation of edges as many as possible.

PC Algorithm:

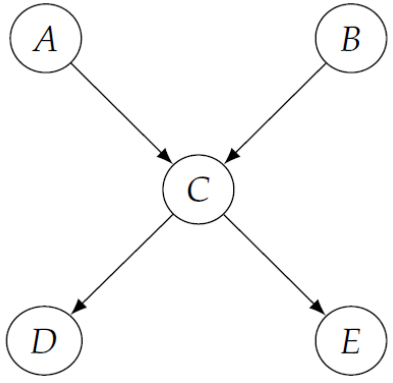
1. Identify the skeleton.
2. Identify immoralities and orient them.
3. Orient qualifying edges that are incident on v-structures.



(1)



(2)



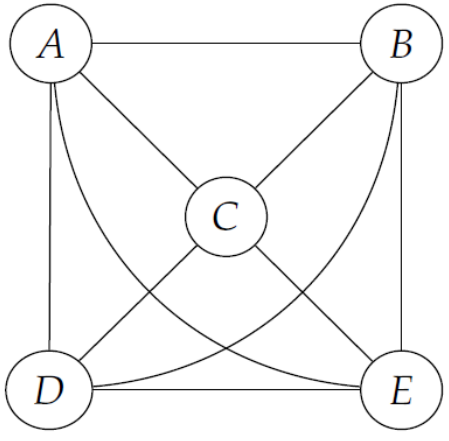
(3)

Identify the Undirected Skeleton

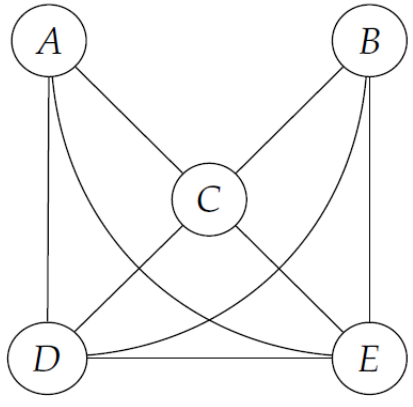
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- Key ideas:
  - If $X_i - X_j$, or $X_i \to X_j$ in a p-Map G, then X_i and X_j are not independent given any variable set U.
  - **Witness**: If X_i and X_j are not adjacent in G, we can find a variable set U that $X_i \perp X_j | U$, denoted as $P \models X_i \perp X_j | U$. U is referred to as a witness of their independence.
- Key steps:

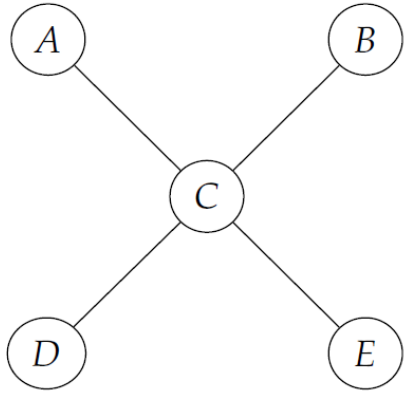
1) Start with a complete graph.
2) Remove an edge $X_i - X_j$, if $X_i \perp X_j | U$
3) Grow the size of U till the size d.



(a) Complete undirected graph that we start with



(b) Undirected graph that remains after removing $X - Y$ edges where $X \perp\!\!\!\perp Y$



(c) Undirected graph that remains after removing $X - Y$ edges where $X \perp\!\!\!\perp Y \mid Z$

Identify the Undirected Skeleton

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- Key ideas:
  - If $X_i - X_j$, or $X_i \rightarrow X_j$ in a p-Map G, then X_i and X_j are not independent given any variable set U.
  - **Witness**: If X_i and X_j are not adjacent in G, we can find a variable set U that $X_i \perp X_j | U$, denoted as $P \models X_i \perp X_j | U$. U is referred to as a witness of their independence.

Procedure Build-PMap-Skeleton (
  $\mathcal{X} = \{X_1, \ldots, X_n\}$, // Set of random variables
  P, // Distribution over $\mathcal{X}$
  d // Bound on witness set
)

1 Let $\mathcal{H}$ be the complete undirected graph over $\mathcal{X}$
2 **for** X_i, X_j in $\mathcal{X}$
3 $\boldsymbol{U}_{X_i,X_j} \leftarrow \emptyset$
4 **for** $\boldsymbol{U} \in$ Witnesses$(X_i, X_j, \mathcal{H}, d)$
5 // Consider $\boldsymbol{U}$ as a witness set for X_i, X_j
6 **if** $P \models (X_i \perp X_j \mid \boldsymbol{U})$ **then**
7 $\boldsymbol{U}_{X_i,X_j} \leftarrow \boldsymbol{U}$
8 Remove X_i—X_j from $\mathcal{H}$
9 **break**
10 **return** $(\mathcal{H}, \{\boldsymbol{U}_{X_i,X_j} : i, j \in \{1, \ldots, n\})$

Time complexity: $O(n^{d+2})$

Skeleton Phase: Reading a Table of p-values

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Cancer network, one million samples, χ^2 tests. Start from the complete graph on the five variables. Test each adjacent pair with conditioning sets of size 0, then 1, then 2, …, drawn from the current neighbours of the pair. Remove the edge the first time the test does **not** reject independence ($p \geq \alpha$) and record that conditioning set as the separating set.

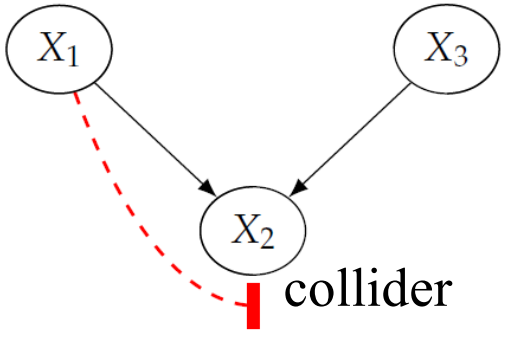
Test	p-value	Decision at $\alpha = 0.01$	Separating set
Pollution ⊥ Smoker	0.035	$p \geq \alpha$: independent, remove edge	∅
every other order-0 test	< 0.001	dependent, keep edge	–
Pollution ⊥ Xray | Cancer	0.668	remove edge	{Cancer}
Pollution ⊥ Dyspnoea | Cancer	0.216	remove edge	{Cancer}
Smoker ⊥ Xray | Cancer	0.440	remove edge	{Cancer}
Smoker ⊥ Dyspnoea | Cancer	0.123	remove edge	{Cancer}
Xray ⊥ Dyspnoea | Cancer	0.249	remove edge	{Cancer}
every other order-1 test	< 0.001	dependent, keep edge	–

Result: skeleton Cancer – Pollution, Cancer – Smoker, Cancer – Xray, Cancer – Dyspnoea (4 of the 10 possible edges).

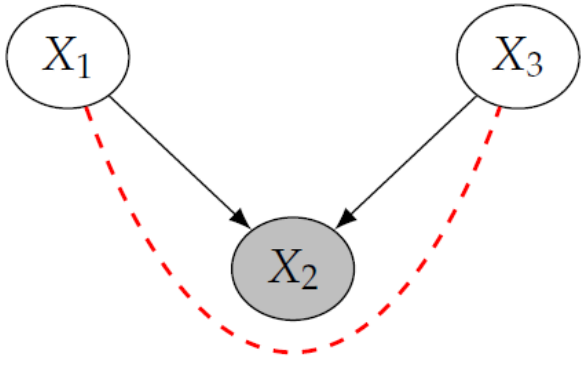
Order 2 is still attempted but accepts no further independence, so the phase ends. **At $\alpha = 0.05$ the first row would be dependent ($0.035 < 0.05$) and the spurious edge Pollution – Smoker would survive.**

Immoralities

- Dependencies:
  - blocked path without condition.
  - $X_1 \perp\!\!\!\perp X_3 | \emptyset$



- Conditional independencies:
  - Unblock path by conditioning on X_2.
  - $X_1 \not\perp\!\!\!\perp X_3 | X_2$

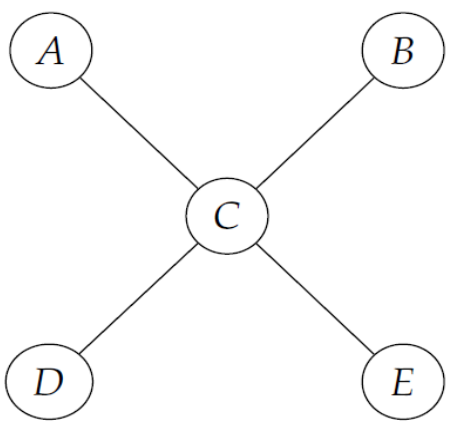


Identify Immoralities

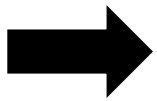
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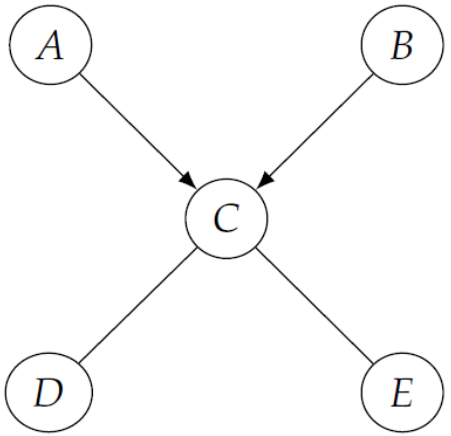
Key steps:

1. Identify the skeleton.
2. Identify immoralities and orient them.
   a. Potential immoralities: $X_i - Z - X_j$ in G, and there is no edge between X_i and X_j .
   b. If ***NOT*** $X_i \perp X_j \mid U$ for any set U that does not contain Z, an immorality is found.
3. Orient qualifying edges that are incident on colliders (v-structure).



skeleton





Find one immorality

Procedure to Find Immoralities

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Procedure Mark-Immoralities (
$\mathcal{X} = \{X_1, \ldots, X_n\}$,
S // Skeleton
$\{\boldsymbol{U}_{X_i,X_j} : 1 \leq i, j \leq n\}$ // Witnesses found by Build-PMap-Skeleton
)

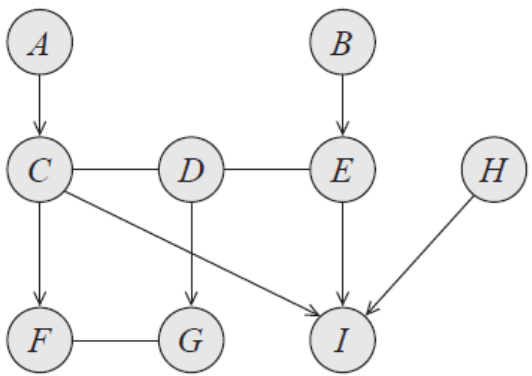
1 $\mathcal{K} \leftarrow S$
2 **for** X_i, X_j, X_k such that $X_i — X_j — X_k \in S$ and $X_i — X_k \notin S$
3 // $X_i — X_j — X_k$ is a potential immorality
4 **if** $X_j \notin \boldsymbol{U}_{X_i,X_k}$ **then**
5 Add the orientations $X_i \rightarrow X_j$ and $X_j \leftarrow X_k$ to $\mathcal{K}$
6 **return** $\mathcal{K}$

Partially Directed Acyclic Graph (PDAG)

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- We need a compact representation of the equivalence class:
  - Find an equivalence class sharing a set of directed edges, other than immoralities.
  - Represent both directed and undirected edges.
- **PDAG**: Let G be a PDAG over $\mathcal{X}$. Let $K_1, \ldots, K_l$ be a disjoint partition of $\mathcal{X}$ such that:
  - The subgraph over K_i contain no directed edges;
  - For any pair of nodes $X_m \in K_i$ and $\mathrm{X_n} \in K_j$, for $i < j$, an edge between X_m and X_n can only be $X_m \rightarrow X_n$.
- Each component K_i is called a chain component.
- A PDAG K is in the equivalence class of a DAG G if it
  - shares the same skeleton as G, and
  - contains a directed edge $X \rightarrow Y$ if and only if

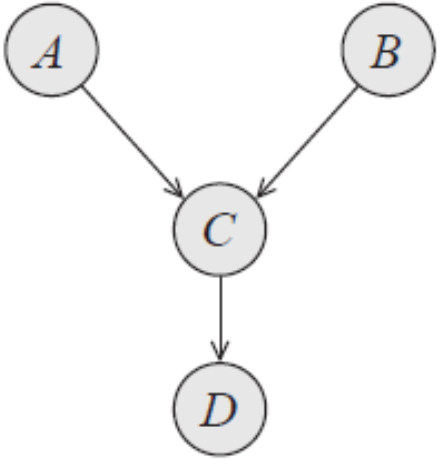
  all G' are I-equivalent to G contain the edge $X \rightarrow Y$.



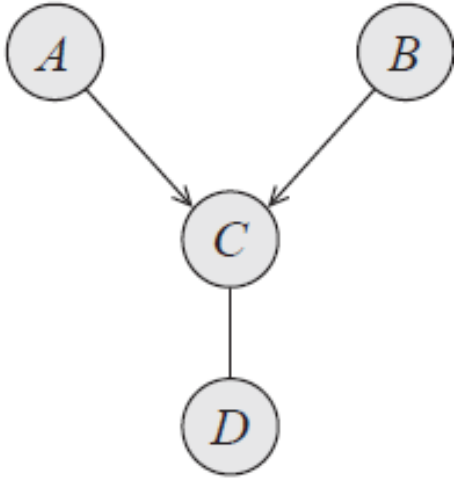
Orient Qualifying Edges

Key steps:

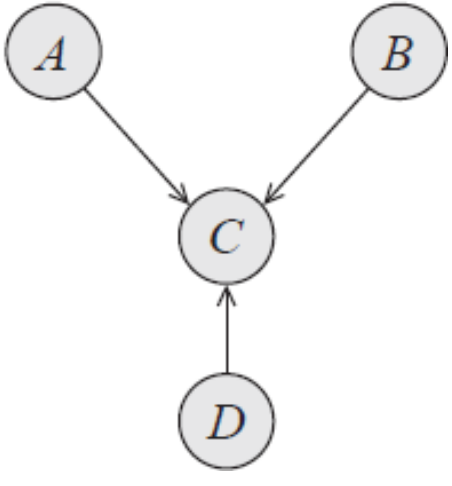
1. Identify the skeleton.
2. Identify immoralities and orient them.
3. Orient qualifying edges that are incident on colliders (v-structures).



(a) Original DAG.



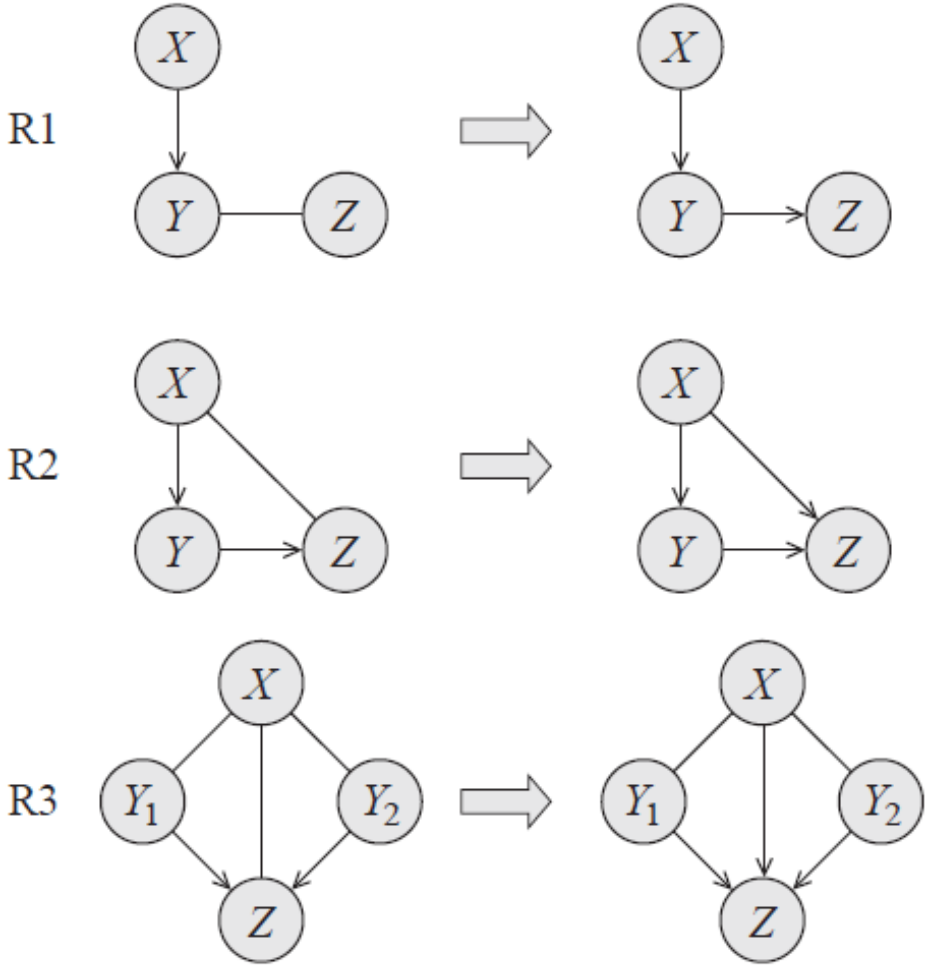
(b) Skeleton with immoralities.



(c) A DAG that is not I-equivalent to (a) and (b).

Orient Qualifying Edges

- **Rule 1**: $Y \rightarrow Z$ is the only option, otherwise a new immorality is included.
- **Rule 2**: X $\rightarrow Z$ is the only option, otherwise a cycle is created.
- **Rule 3**: By contradiction, if X $\leftarrow Z$, we must have $Y_1 \rightarrow X$, and $Y_2 \rightarrow X$ to avoid creating cycles. As a result, $Y_1 \rightarrow X \leftarrow Y_2$ forms an immorality, contradicts the fact that there is no immorality between $Y_1 - X - Y_2$ in the PDAG.
- Iterative application of Rule 1 to Rule 3 may terminate at some points.



Final Algorithm

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Procedure Build-PDAG (
  $\mathcal{X} = \{X_1, \ldots, X_n\}$ // A specification of the random variables
  P // Distribution of interest
)

1 $S, \{U_{X_i, X_j}\} \leftarrow$ Build-PMap-Skeleton$(\mathcal{X}, P)$
2 $\mathcal{K} \leftarrow$ Find-Immoralities$(\mathcal{X}, S, \{U_{X_i, X_j}\})$
3 **while** not converged
4 Find a subgraph in $\mathcal{K}$ matching the left-hand side of a rule R1–R3
5 Replace the subgraph with the right-hand side of the rule
6 **return** K

What PC Returns: a CPDAG

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What PC returns

- A **PDAG** with directed edges and undirected edges.
- The **CPDAG** (completed PDAG) is the unique representation of an I-equivalence class: same skeleton, same immoralities, and every edge that rules R1–R3 can orient is oriented.
- Any DAG of the class is obtained by orienting the undirected edges without creating a new v-structure or a cycle.

Consequences

- Even with perfect independence tests, PC recovers the CPDAG of the true DAG, **not the DAG**: independence tests cannot tell I-equivalent DAGs apart.
- $X \rightarrow Y \rightarrow Z$, $X \leftarrow Y \leftarrow Z$ and $X \leftarrow Y \rightarrow Z$ are I-equivalent, so PC returns $X - Y - Z$; only the v-structure $X \rightarrow Y \leftarrow Z$ is identifiable on its own.

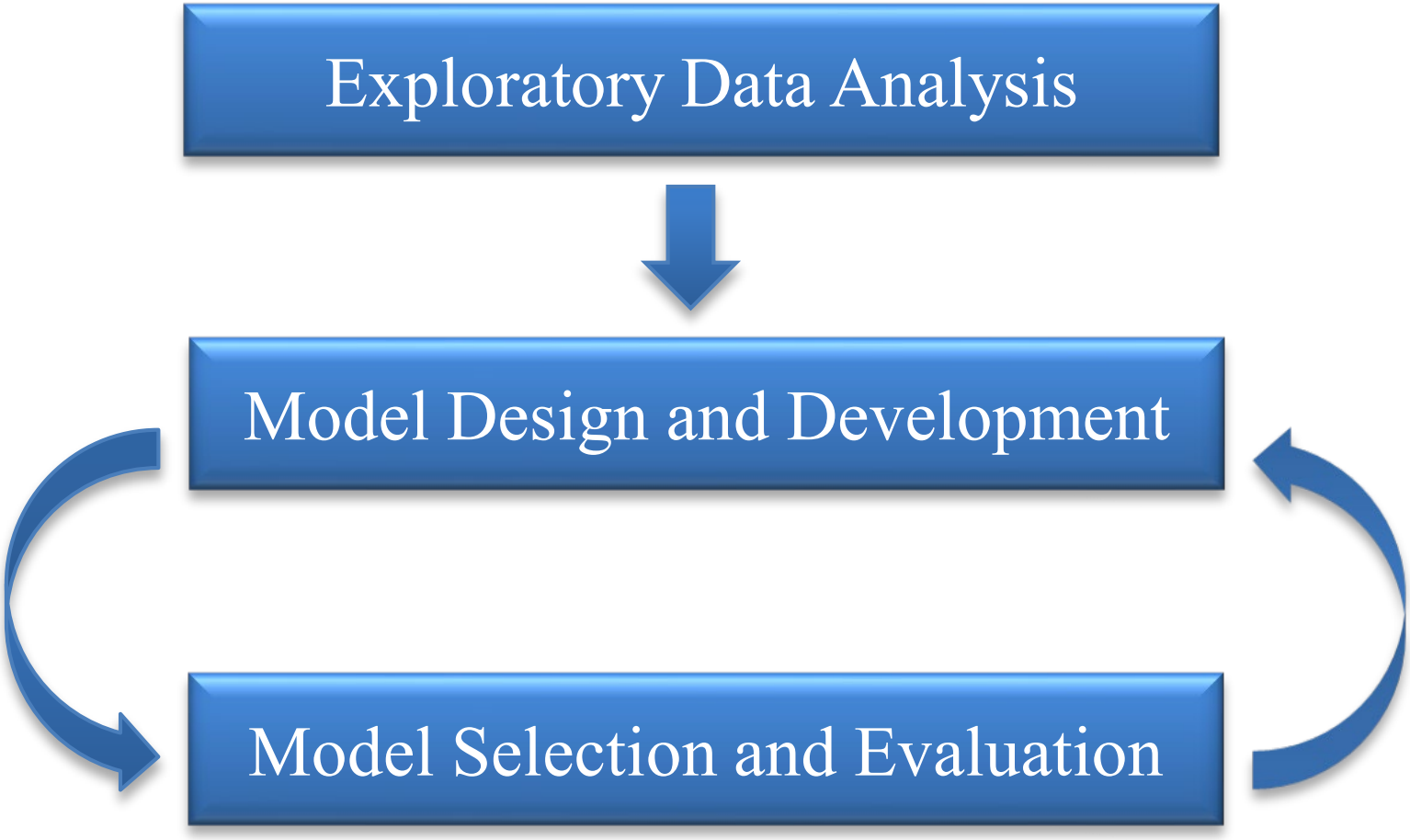
Activity I&II: PC Algorithm

- Which CI test or significance level leads to a better DAG?
- Identification of immoralities.

Summary

- What we learned:
  - PC algorithm.
  - Structure learning in practice.
- Relevant readings:
  - Sections 3.4 and 18.2 of the book “Probabilistic Graphical Models, Principles and Techniques”.
- Acknowledgement
  - Some figures used in these slides are from the book “Probabilistic Graphical Models, Principles and Techniques”, and Pgmpy library.

Overview of the Unit



Model Design and Development

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Model User

- Lasso
- Ridge
- Subset selection
- Regression splines
- Smooth splines
- Neural networks
- PCA
- t-SNE
- k-Means
- Agglomerative Clustering

Model Designer

- Bayesian Networks
- Markov Networks
- Sum-Product
- Max-Product
- Forward Sampling
- Rejection Sampling
- PC
- Gibbs Sampling

Final Exam

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- Closed book with specially permitted items.
  - Notes: double sided A4 x 5 Max – Physical only.
  - Physical / virtual calculator
- Similar to the sample/mock exam.
- Do not need to write code in the final exam, but you should be able to analyze the results of the code.

